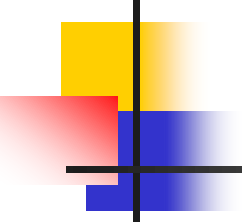
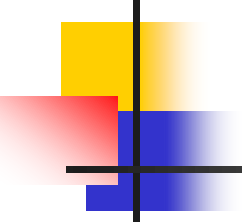
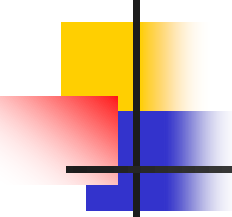


Software Engineering

- 
-
- The notion of **software engineering** was introduced in 1968 at a conference held to discuss what was then called the 'software crisis', resulted from the first attempts to develop large sized software systems.
 - However, after more than 50 years **the software crisis still exists ...**

- 
- The IEEE gives the following definition [IEEE90]:
 - **Software Engineering (SE)** is:
 1. *The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is the application of engineering to software.*
 2. *The study of approaches as in 1.*
 - SE is mainly concerned with the development of **large sized software systems**.
 - SE employs specific techniques, methods and tools to control the complexity inherent in large software systems.



Main References and Recommended Reading

- [Som15] I. Sommerville, *Software Engineering*, (10th edition), Addison-Wesley, 2016 (and previous editions).
- [LL05] T. Lethbridge and R. Laganier, *Object-Oriented Software Engineering: Practical Software Development using UML and Java* (2nd edition), McGraw-Hill, 2005. <http://www.site.uottawa.ca/school/research/lloseng/>
- [FP20] A. Fox and D. Patterson, *Engineering Software as a Service: An Agile Approach using Cloud Computing*, (1st, 2nd editions) Strawberry Canyon, (2016, 2020). <http://www.saasbook.info/>
- [GoF94] E. Gamma, R. Helm, R. Johnson and J. Vlissides, *Design Patterns: Elements of Reusable Object-Oriented Software*, Addison Wesley, 1994.
- [CHVB18] E.M. Clarke, T.A. Henzinger, H. Veith and R. Bloem, Editors, *Handbook of Model Checking*, Springer, 2018.
- [Pie13] B. Pierce et al, *Software Foundations*, 2020. <https://softwarefoundations.cis.upenn.edu/>
- [IEEE90] *IEEE Standard Glossary of Software Engineering Terminology*, IEEE Standard 610.12-1990, 1990.